MINISTRY OF DEFENCE (DEPT. OF DEFENCE PRODUCTION) • RESTRICTED — DEFENCE PRODUCTION

STANDARD OPERATING PROCEDURE: HIGH-POWER MULTI-FUEL V12 ENGINE DROPPING & ALIGNMENT

0.1 DOCUMENT CONTROL & METADATA MATRIX

DOCUMENT ID	**HVF/AVNL/SOP-MBT/EA-309**	**REVISION & DATE**	Rev 5.1 (2026) | 30-Oct-2026
ISSUING AUTHORITY	Heavy Vehicles Factory (HVF), Avadi	**CORPORATE ENTITY**	Armoured Vehicles Nigam Ltd (AVNL)
TARGET PLATFORM	MBT T-90S (V-92S2 1000 HP) / CVRDE 1500 HP Powerpack	**PRODUCTION SHOP**	Powerpack Integration & Dynamometer Test Cell (Shop-12)
QA SURVEILLANCE	DGQA (Armoured Vehicle Electronics & Hull)	**PAGE BUDGET**	**8 Pages (Controlled Distribution)**
APPLICABLE STD	ISO 9001:2015 / GOST-V / MIL-STD-1949	**SECURITY CLASS**	**RESTRICTED — DEFENCE PRODUCTION**

0.2 EXECUTIVE SUMMARY & OPERATIONAL MANDATE

This Standard Operating Procedure (SOP) defines the engineering sequence for lowering, trunnion-mounting, laser-aligning, and fluid/electrical coupling of the 1,000–1,500 HP turbocharged V12 multi-fuel diesel engine and planetary transmission powerpack into the MBT rear hull compartment at HVF Avadi. Strict coaxial alignment between the engine output damper and gearbox input shaft prevents spline fretting and torsional vibration failure under severe cross-country combat loads.

This controlled technical procedure applies to all CNC operators, defence metallurgists, millwright technicians, and DGQA resident inspectors assigned to **Powerpack Integration & Dynamometer Test Cell (Shop-12)** at the HVF Avadi manufacturing complex. Autem in blanditiis voluptatum tenetur recusandae asperiores nihil quasi harum. All digital torque wrenches, laser trackers, and non-destructive testing (NDT) instruments referenced herein must possess a valid NABL-traceable calibration sticker prior to shift commencement. Ipsam rerum quasi quidem similique illum fuga perspiciatis recusandae deleniti.

0.3 CRITICAL ENGINEERING ACCEPTANCE BENCHMARKS

CRITICAL PARAMETER	MANDATORY ENGINEERING ACCEPTANCE CRITERION
Shaft Coaxiality	Radial offset ≤ 0.05 mm; angular misalignment ≤ 0.04 mm/100 mm
Mount Load Balance	Trunnion & resilient pad load distribution within ± 3.5% design mass
Cooling Loop Seal	Zero pressure decay at 2.2 bar pneumatic/hydrostatic test for 30 min
Hot-Cell Test Run	Full rated power (736–1,103 kW) @ 2,000 RPM; oil pressure 4.5–6.0 bar

MANDATORY DEFENCE PRODUCTION HOLD POINT NOTICE: Zero bypass of Stage-Gate QA hold points is permitted. Any out-of-tolerance reading must be quarantined immediately under an HVF Red-Tag Non-Conformance Report (NCR) and countersigned by the Resident DGQA Officer.

SECTION 1.0 — ENGINE BAY HULL PEDESTAL INSPECTION & RESILIENT MOUNT PRE-CALIBRATION

Operational Execution Protocol: Before powerpack lowering, the rear hull engine bay floor, machined side trunnion brackets, and final-drive bulkhead bores are cleaned, dimensionally surveyed, and fitted with pre-measured shim packs for the four main mounting pads. During shop-floor execution of *Engine Bay Hull Pedestal Inspection & Resilient Mount Pre-Calibration*, the designated chargehand shall verify that calibration certificate **CAL-HVF-4013** and material traceability record **LOT-KW-9235** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, impedit corrupti odit voluptates cumque id aliquid, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table Engine must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, iure expedita quos consequuntur corrupti mollitia placeat hic nihil, while automated telemetry from station **CAL-HVF-5719** is archived against hull serial dossier **LOT-WL-7533** prior to releasing the assembly to the subsequent production workstation.

1.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **1.0** on platform *MBT T-90S (V-92S2 1000 HP) / CVRDE 1500 HP Powerpack*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Numquam enim voluptatem modi voluptatibus exercitationem vel quaerat labore ullam. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 1.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
1.1	Mounting Pedestal Co-Planarity	**≤ 0.08 mm across 4 hull pads**	Laser Level & Optical Prism	Pre-Drop Survey
1.2	Final Drive Bore Coaxiality	**≤ 0.06 mm port-to-starboard**	Laser Bore Alignment Kit	100% Hull Bays
1.3	Elastomeric/Solid Pad Stiffness	**Within ± 4% rated kN/mm curve**	Hydraulic Compression Bench	Matched Set of 4
1.4	Bilge Scavenge Valve Operation	**≥ 65 L/min evacuation flow**	Flowmeter Functional Check	Pre-Drop Verification
1.5	Fire-Wire Thermal Loop Continuity	**24.0 Ω ± 0.5 Ω loop resistance**	AFDSS Test Harness Box	Dual Zone Loops

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-424	SHIFT-A / EA-309	OP-761 (Verified)	DGQA-34 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 1.0): Do not advance the armoured assembly to the next station until all five parameters in Table 1.0 are counter-stamped in the hull e-Passport ledger.

SECTION 2.0 — V12 POWERPACK HOISTING, TRUNNION LOWERING & SHIM PACK SELECTION

Operational Execution Protocol: The dressed V12 multi-fuel engine and hydraulic-mechanical transmission sub-frame are hoisted using the calibrated 5-tonne four-point turnbuckle rig, lowered through the deck opening, and seated onto the hull mounting trunnions. During shop-floor execution of *V12 Powerpack Hoisting, Trunnion Lowering & Shim Pack Selection*, the designated chargehand shall verify that calibration certificate **CAL-HVF-8658** and material traceability record **LOT-QF-3051** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, ea perspiciatis impedit laboriosam eligendi consectetur velit tenetur aliquid, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table V12 must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, odio suscipit ratione inventore ut amet nisi, while automated telemetry from station **CAL-HVF-5141** is archived against hull serial dossier **LOT-HT-7276** prior to releasing the assembly to the subsequent production workstation.

2.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **2.0** on platform *MBT T-90S (V-92S2 1000 HP) / CVRDE 1500 HP Powerpack*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. In officia eius distinctio veritatis ab alias voluptas labore cupiditate temporibus veniam. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 2.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
2.1	Hoist Sling Load Symmetry	**± 2.5% across 4 crane load cells**	Wireless Shackle Load Cells	Continuous Lift
2.2	Deck Clearance Envelope	**≥ 18.0 mm minimum wall clearance**	Go/No-Go Clearance Wand	4 Hull Corners
2.3	Pad Seating Unclamped Gap	**≤ 0.04 mm after shim insertion**	Calibrated Stainless Feeler	All 4 Mount Pads
2.4	Primary Mount Bolt Torque	**520 Nm ± 15 Nm + 60° angle turn**	Electronic Torque-Angle Tool	16 Main Anchor Bolts
2.5	Chassis Ground Bonding Strap	**≤ 1.2 mΩ hull-to-crankcase**	Kelvin Bridge Micro-Ohmmeter	Dual Braided Straps

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-239	SHIFT-A / EA-309	OP-424 (Verified)	DGQA-64 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 2.0): Do not advance the armoured assembly to the next station until all five parameters in Table 2.0 are counter-stamped in the hull e-Passport ledger.

SECTION 3.0 — DUAL-LASER CRANKSHAFT-TO-TRANSMISSION INPUT SHAFT COAXIAL ALIGNMENT

Operational Execution Protocol: A dual-laser shaft alignment system is clamped to the V12 flywheel torsional damper flange and the transmission input hub. Horizontal and vertical shimming compensates for cold-to-hot thermal growth (+0.12 mm vertical rise at 95°C). During shop-floor execution of *Dual-Laser Crankshaft-to-Transmission Input Shaft Coaxial Alignment*, the designated chargehand shall verify that calibration certificate **CAL-HVF-0441** and material traceability record **LOT-DD-3496** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, reprehenderit nam dicta id sed soluta non qui nemo provident laudantium quo voluptatem, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table Dual-Laser must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, repellat totam veniam eaque perferendis accusamus eligendi cumque ut quod eius, while automated telemetry from station **CAL-HVF-4602** is archived against hull serial dossier **LOT-RZ-8182** prior to releasing the assembly to the subsequent production workstation.

3.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **3.0** on platform *MBT T-90S (V-92S2 1000 HP) / CVRDE 1500 HP Powerpack*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Vitae nulla illo nam ab nam architecto sunt voluptatem aliquam et consectetur accusantium amet accusantium. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 3.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
3.1	Parallel Radial Offset (Cold)	**≤ 0.05 mm TIR (target -0.12 mm V)**	Dual-Laser Shaft Aligner	9-12-3-6 O'Clock Sweep
3.2	Angular Face Misalignment	**≤ 0.04 mm per 100 mm diameter**	Dual-Laser Shaft Aligner	Continuous 360° Sweep
3.3	Gear Coupling Axial Float	**1.50 mm – 2.20 mm axial end-play**	Magnetic Base Dial Gauge	Post-Coupling Check
3.4	Final Drive Cardan Spline Runout	**≤ 0.07 mm radial runout**	Dial Indicator on Yoke Hub	Port & Starboard Drives
3.5	Torsional Damper Fastener Torque	**195 Nm ± 6 Nm (Loctite 262)**	Calibrated Click/Digital Wrench	All 12 Coupling Bolts

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-170	SHIFT-A / EA-309	OP-392 (Verified)	DGQA-73 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 3.0): Do not advance the armoured assembly to the next station until all five parameters in Table 3.0 are counter-stamped in the hull e-Passport ledger.

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SECTION 4.0 — HIGH-PRESSURE FUEL, DRY-SUMP LUBRICATION & COOLING LOOP COUPLING

Operational Execution Protocol: High-pressure braided PTFE fuel lines, dry-sump scavenging oil pipes, and ethylene-glycol radiator manifolds are connected via quick-disconnect self-sealing couplings and subjected to hydrostatic and pneumatic pressure-hold testing. During shop-floor execution of *High-Pressure Fuel, Dry-Sump Lubrication & Cooling Loop Coupling*, the designated chargehand shall verify that calibration certificate **CAL-HVF-7167** and material traceability record **LOT-YM-0042** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, aliquam iste dicta est voluptates labore possimus possimus atque in, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table High-Pressure must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, provident quos quod similique ipsum accusantium excepturi libero quam nesciunt nulla dignissimos sed, while automated telemetry from station **CAL-HVF-7186** is archived against hull serial dossier **LOT-DI-1337** prior to releasing the assembly to the subsequent production workstation.

4.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **4.0** on platform *MBT T-90S (V-92S2 1000 HP) / CVRDE 1500 HP Powerpack*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Facilis soluta consequatur nesciunt soluta nobis totam voluptates unde pariatur. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 4.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
4.1	Fuel Supply & Return Proof Test	**6.0 bar hydrostatic for 20 min**	Digital Pressure Recorder	Zero Weepage Allowed
4.2	Dry-Sump Oil Circuit Flush	**NAS 1638 Class 6 cleanliness**	Laser Particle Counter	15-Min Bypass Flush
4.3	Coolant System Pressure Hold	**2.2 bar pneumatic for 30 min**	Differential Decay Gauge	≤ 0.02 bar Drop
4.4	High-Pressure Air Start Bottle	**150 bar charge; zero leak soap test**	Calibrated Bourdon/Digital	Dual Air Cylinders
4.5	Hose Clamp Torque & Fire Sleeve	**8.5 Nm ± 0.5 Nm on twin clamps**	Calibrated Torque Screwdriver	100% Fluid Joints

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-755	SHIFT-A / EA-309	OP-485 (Verified)	DGQA-06 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 4.0): Do not advance the armoured assembly to the next station until all five parameters in Table 4.0 are counter-stamped in the hull e-Passport ledger.

SECTION 5.0 — CYCLONIC AIR FILTRATION, TURBOCHARGER DUCTING & EXHAUST SEALING

Operational Execution Protocol: Two-stage cyclonic dust-ejection air cleaner cassettes, compressor intake V-band bellows, and Inconel exhaust ejector manifolds are sealed and vacuum-tested to prevent desert silica ingestion and deep-fording water ingress. During shop-floor execution of *Cyclonic Air Filtration, Turbocharger Ducting & Exhaust Sealing*, the designated chargehand shall verify that calibration certificate **CAL-HVF-9092** and material traceability record **LOT-ZK-6628** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, voluptas dolorem ex hic impedit saepe corporis temporibus et iure ducimus quam consequuntur, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table Cyclonic must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, consequatur cum dolore iusto voluptate quos placeat nisi praesentium aliquid, while automated telemetry from station **CAL-HVF-8704** is archived against hull serial dossier **LOT-LQ-0188** prior to releasing the assembly to the subsequent production workstation.

5.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **5.0** on platform *MBT T-90S (V-92S2 1000 HP) / CVRDE 1500 HP Powerpack*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Est iure modi reprehenderit doloremque quibusdam corporis sint fugit facilis. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 5.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
5.1	Intake Tract Vacuum Decay	**Hold -15 kPa for 120 seconds**	Manometer Vacuum Rig	Compressor Inlet Seal
5.2	Cyclone Dust Scavenge Ejector	**≥ 98.5% separation efficiency**	Differential Pressure Sensor	Exhaust Ejector Port
5.3	Exhaust Manifold V-Band Torque	**28 Nm ± 2 Nm with anti-seize**	Calibrated Torque Wrench	All 8 V-Band Clamps
5.4	Deep-Fording Exhaust Valve Seal	**Zero bubble leak at 5.0 m head**	Hydrostatic Back-Pressure Rig	Snorkel Mode Interlock
5.5	Thermal Insulation Blanket Temp	**Surface ≤ 140°C at full load**	IR Thermal Imaging Camera	Exhaust Shroud Survey

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-637	SHIFT-A / EA-309	OP-827 (Verified)	DGQA-60 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 5.0): Do not advance the armoured assembly to the next station until all five parameters in Table 5.0 are counter-stamped in the hull e-Passport ledger.

SECTION 6.0 — COLD MOTORING, HOT-RUN HULL TELEMETRY & VIBRATION SIGNATURE SIGN-OFF

Operational Execution Protocol: With pre-lube oil pressure established via the electric priming pump, the V12 engine is motored, started, and stepped through idle, mid-range, and full-load stall tests while triaxial accelerometers record hull vibration velocity. During shop-floor execution of *Cold Motoring, Hot-Run Hull Telemetry & Vibration Signature Sign-Off*, the designated chargehand shall verify that calibration certificate **CAL-HVF-7112** and material traceability record **LOT-EU-0637** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, consequatur porro nihil occaecati cupiditate deleniti vero illo iste accusantium consectetur molestiae voluptatum porro hic, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table Cold must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, autem asperiores deserunt occaecati ad debitis a autem sapiente, while automated telemetry from station **CAL-HVF-2941** is archived against hull serial dossier **LOT-RI-0511** prior to releasing the assembly to the subsequent production workstation.

6.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **6.0** on platform *MBT T-90S (V-92S2 1000 HP) / CVRDE 1500 HP Powerpack*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Voluptates enim necessitatibus similique consequatur distinctio magni qui aperiam dolorem reiciendis quasi temporibus consectetur cupiditate dolor repellat. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 6.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
6.1	Pre-Start Oil Priming Pressure	**≥ 1.8 bar within 12 seconds**	Cockpit & CAN-Bus Transducer	Mandatory Start Interlock
6.2	Rated Speed & Oil Pressure	**2,000 RPM / 4.5 – 6.0 bar @ 90°C**	Engine Control Telemetry	30-Min Sustained Run
6.3	Turbo Boost & Exhaust Gas Temp	**2.4 bar abs boost / EGT ≤ 680°C**	K-Type Thermocouple Array	Full Stall Load Test
6.4	Triaxial Vibration Velocity	**≤ 4.5 mm/s RMS (ISO 10816-6)**	FFT Vibration Analyzer	All 4 Engine Mounts
6.5	Hot-State Laser Re-Alignment	**≤ 0.04 mm radial offset at 95°C**	Dual-Laser Hot Verification	Immediate Post-Shutdown

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-863	SHIFT-A / EA-309	OP-787 (Verified)	DGQA-76 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 6.0): Do not advance the armoured assembly to the next station until all five parameters in Table 6.0 are counter-stamped in the hull e-Passport ledger.

APPENDIX A — NORMATIVE DEFENCE STANDARDS, CALIBRATION & REVISION LEDGER

This Appendix establishes the normative statutory, metallurgical, and metrological references governing **High-Power Multi-Fuel V12 Engine Dropping & Alignment** (HVF/AVNL/SOP-MBT/EA-309) at Heavy Vehicles Factory (HVF), Avadi. All instrumentation utilized in Sections 1.0 through 6.0 must maintain unbroken calibration traceability to National Physical Laboratory (NPL) / NABL standards. Enim accusamus esse cupiditate sapiente maxime eveniet perspiciatis.

STD / CODE	STANDARD TITLE & DEFENCE MANUFACTURING APPLICABILITY	CUSTODIAN
DGAQA / DGQA-ARM-09	Quality Assurance Norms for Combat Vehicle Hull, Turret & Running Gear	DGQA (MoD)
MIL-DTL-12560 / DMR	Armor Plate, Steel, Wrought, Homogeneous & High-Hardness Ballistic Spec	DMRL / HVF Met Lab
ISO 10012 / NABL-141	Measurement Management Systems — Metrological Confirmation & Calibration	HVF Metrology Cell
AVNL-HSE-SOP-014	Industrial Safety, Crane Rigging, Fume Extraction & Confined Space Entry	HVF Safety Dept

APPENDIX B — MULTI-TIER OFFICIAL SIGN-OFF & AUTHORIZATION MATRIX

This Standard Operating Procedure is officially released for defence production upon execution of the four-tier authorization matrix below by designated HVF, AVNL, and DGQA authorities at Avadi, Chennai.

TIER / ROLE	DESIGNATION & DIRECTORATE	OFFICER NAME & ID	DIGITAL STAMP / SIGNATURE	DATE
TIER 1: PREPARED BY	Deputy General Manager (Production Engg) HVF Avadi	Er. R. Venkataraman IOFS-4418	**[DIGITALLY SIGNED] DGM/PE/HVF-AVADI**	30-Oct-2026
TIER 2: REVIEWED BY	Joint General Manager (Quality & Metallurgy) HVF Avadi	Dr. S. Muthukumar IOFS-3802	**[DIGITALLY SIGNED] JGM/QA-MET/HVF**	30-Oct-2026
TIER 3: CONCURRENCE	Controller, Directorate General of Quality Assurance (DGQA Armoured Vehicles)	Brig. A. K. Rathore CQA(HV) Avadi	**[SEAL & COUNTERSIGNED] CQA(HV)/MOD/GOI**	30-Oct-2026
TIER 4: APPROVED BY	Chief General Manager / Executive Director Heavy Vehicles Factory (AVNL)	Shri K. Parthasarathy CGM / HVF Avadi	**[APPROVED FOR RELEASE] CGM-OFFICE/AVNL**	30-Oct-2026

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